System Performance Evaluation

Functionality, efficiency and usability in the context of civil registry document digitization

Objective
Evaluate the overall system performance in terms of functionality, efficiency, and usability in the context of civil registry document digitization.

1. What this objective means

After the system is built, it is tested and judged on three qualities. The test uses civil registry documents (birth, marriage, death) and the people who would really use the system.

Quality	The question it answers	How it is measured	Part of this file
Functionality	Does the system do what it is supposed to do?	Feature test cases marked Pass or Fail; OCR reading accuracy.	Part 3
Efficiency	Is it faster and less work than typing the records by hand?	Timing tests: system vs. manual encoding; number of corrections.	Part 4
Usability	Is it easy to learn and use for the registry staff?	Questionnaire given to users after they try the system (rating scale and SUS).	Part 5

2. Evaluation plan

Item	Plan (fill in or confirm with your adviser)
Respondents	Civil registry staff who will use the system, plus IT or subject experts if the panel wants them. Number of respondents: _____
Test documents	A set of civil registry documents (birth, marriage, death) of different handwriting and scan quality. Number of documents: _____. Use copies with personal data removed or approved for testing.
Procedure	1) Short orientation. 2) Each respondent performs the tasks in Part 3 and times the digitization in Part 4. 3) Each respondent answers the questionnaire in Part 5. 4) Results are tallied.
Standard used	ISO/IEC 25010 software quality model (functional suitability, performance efficiency, usability) and the System Usability Scale (SUS). Confirm the standard your school expects.
Where it goes in the manuscript	Methodology chapter: this plan and the instruments. Results chapter: the tallied tables. Conclusion: the answer to the objective.

Part 3. Functionality test checklist

Tester performs each task and marks Pass or Fail. Record what happened in Remarks if it failed.

#	Task / feature	Expected result	Pass	Fail	Remarks
1	Log in with a staff account and change the password	Account opens the dashboard; password change is saved.			
2	Upload a scanned civil registry page	The page appears in the workspace and is stored.			
3	Choose a document type and template	The correct template layout is applied to the page.			
4	Align the page and snap rows to the lines	Row lines follow the table rows, even on a slightly tilted scan.			
5	Run the text reading (OCR) on the page	Each line is read and the text appears in the matching field.			
6	Verify and correct the read text	Staff can edit a wrong value and save the corrected record.			
7	Add a missed field manually	A box drawn on the page is read and added to the record.			
8	Save the record to the database	The record is stored with its scanned page image.			
9	Search and open a saved record	Records can be listed, searched and opened with the scan beside the data.			
10	Request a change to a saved record	A change request is created and an approver can approve or reject it.			
11	View the dashboard and reports	Counts and charts match the records in the system.			
12	Export reports to CSV	The CSV file opens and contains the correct records.			
13	Manage users (add, deactivate, reset password)	An admin can do all three and the user list updates.			
14	Check the audit log	Actions such as uploads and edits are listed with user and time.			
15	Build or edit a document template	A new template can be created, tested on a sample and activated.			

Functional suitability score = (number of Pass / total tasks) x 100 = _____ %

OCR reading accuracy (optional but recommended). For a sample of lines, compare the text the system read with the correct text typed by a person. Character accuracy = (1 - character error rate) x 100.

Document type	Lines checked	Lines read exactly right	Accuracy (%)
Birth			
Marriage			
Death			

Part 4. Efficiency timing sheet

Use the same documents for both columns. Time one document from the start (blank) until the record is fully saved and correct. Use a stopwatch and record minutes and seconds.

Doc no.	Document type	Manual encoding time	System time (scan to saved)	Corrections made in Verify	Time saved
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
Average					

Time saved (%) = (manual time - system time) / manual time x 100
Throughput = documents completed per hour with the system vs. by hand.

Section B of the questionnaire in Part 5 collects the users' opinion on efficiency. Report it together with these timing results.

Part 5. Respondent questionnaire

Name (optional): _____ Position: _____ Date: _____

Directions: Rate each statement. Encircle one number. **5** Strongly Agree, **4** Agree, **3** Neutral, **2** Disagree, **1** Strongly Disagree.

#	A. Functionality	5	4	3	2	1
1	The system has all the features needed to digitize civil registry documents.					
2	The system reads the text of the scanned documents correctly.					
3	The records saved by the system are complete and accurate.					
4	Searching, viewing and exporting records works as expected.					
5	The change request and approval process works as expected.					

#	B. Efficiency	5	4	3	2	1
6	The system digitizes a document faster than typing it manually.					
7	Correcting the read text takes little effort.					
8	The system responds quickly when I scan, read and save.					
9	The system reduces the paperwork and repeated encoding of records.					
10	The system helps me finish more records in the same amount of time.					

#	C. Usability	5	4	3	2	1
11	The system is easy to learn.					
12	The screens and buttons are clear and easy to understand.					
13	I can finish my tasks without help from another person.					
14	Error and warning messages are clear and tell me what to do.					
15	I would be comfortable using this system every day.					

D. System Usability Scale (SUS)

Same scale: 5 = Strongly Agree ... 1 = Strongly Disagree.

#	SUS statement	5	4	3	2	1
16	I think that I would like to use this system frequently.					
17	I found the system unnecessarily complex.					
18	I thought the system was easy to use.					
19	I think that I would need the support of a technical person to be able to use this system.					
20	I found the various functions in this system were well integrated.					
21	I thought there was too much inconsistency in this system.					
22	I would imagine that most people would learn to use this system very quickly.					
23	I found the system very cumbersome to use.					
24	I felt very confident using the system.					
25	I needed to learn a lot of things before I could get going with this system.					

Comments and suggestions:

_____

_____

_____

_____

Part 6. How to score and interpret

Rating scale (Parts A to C). Compute the mean of each item and of each section. Describe the mean using this table.

Mean range	Verbal interpretation
4.21 - 5.00	Strongly Agree (Excellent)
3.41 - 4.20	Agree (Very Good)
2.61 - 3.40	Neutral (Good)
1.81 - 2.60	Disagree (Fair)
1.00 - 1.80	Strongly Disagree (Poor)

SUS score (Part D). For odd-numbered statements (1, 3, 5, 7, 9) subtract 1 from the rating. For even-numbered statements (2, 4, 6, 8, 10) subtract the rating from 5. Add the ten results and multiply by 2.5. The result is out of 100. A score of about 68 is the average; 80 and above is considered excellent.

Putting the results in the manuscript.

Section	Report
Functionality	Pass rate from Part 3, OCR accuracy table, and the mean of section A.
Efficiency	Average time saved and throughput from Part 4, and the mean of section B.
Usability	Mean of section C and the SUS score from Part D.
Overall	Average of the three section means, then one sentence answering the objective.

Note: the items above are a starting draft. Adjust the wording and the number of items to match what your adviser and panel require.